

Brain Tumor Segmentation and Localization Using Traditional Image Processing and OpenCV

Weekly Status Report Week 4 of 10

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1 Introduction

The objective of the project is to segment and localize the Whole Tumor (WT) region in brain MRI using a traditional approach. It uses image-processing techniques rather than machine-learning or deep-learning models. The current pipeline carries out intensity-based rules on multimodal MRI data to select candidate tumor voxels and produces a binary mask, which is then refined through three-dimensional morphological processing. The predicted masks are evaluated against the ground-truth segmentations provided in the BraTS2020 training dataset. This week's work builds on the study of three classical segmentation approaches and extends the evaluation from an initial single-slice comparison to 20 MRI cases.

2 Objective

The objectives for this reporting period were:

- Review classical thresholding and morphology-based approaches for MRI tumor segmentation.
- Implement an intensity-based segmentation pipeline and produce binary Whole Tumor masks.
- Assess the pipeline on 20 cases with respect to the BraTS2020 ground truth.
- Compute Dice, Intersection over Union (IoU), precision, recall, and boundary/volume metrics to evaluate segmentation quality.
- Determine the key sources of error and identify changes for the subsequent iteration.

3 Literature Review

3.1 Hybridized Multilevel Thresholding and Level Set

Dawngliana et al. present a hybrid method that combines median filtering and histogram equalization with multilevel thresholding based on Type-II fuzzy sets. Morphological processing removes small noisy regions, while a level-set method refines the tumor boundary. The paper reports improved Dice and Jaccard performance compared with its referenced FCM–Level Set approach. This work motivates combining intensity-based candidate selection with a boundary-refinement stage. [1]

3.2 Thresholding and Morphological Operations

Murthy and Sadashivappa describe a comparatively simple pipeline involving Sobel-based preprocessing, histogram equalization, border removal, thresholding, and binary morphological operations such as erosion, dilation, opening, and closing. The resulting tumor region can also be characterized by features such as area, perimeter, and centroid. The method is relevant because it is relatively fast and straightforward to implement. [2]

3.3 Integrated Thresholding and Morphological Processing

Islam, Imteaz, and Marium-E-Jannat use FLAIR MRI and combine image enhancement, global/Otsu thresholding, restoration of nearby tumor pixels, morphological processing, hole filling, and removal of small connected components. Their reported detection rate was 66 out of 76 FLAIR cases. They also note that results can be affected by noise and image clarity. [3]

3.4 Summary

The reviewed studies show that thresholding can identify candidate tumor regions, while morphology and boundary refinement can reduce artifacts and improve mask shape. However, fixed thresholds and empirically selected parameters may not perform equally well across MRI cases. The current project therefore evaluates segmentation quality over multiple cases rather than relying only on a single example.

4 Work Completed This Week

4.1 Methodology

The current implementation uses multimodal MRI volumes and intensity-based rules to identify candidate tumor voxels. The selected voxels are combined into a binary mask, which is refined using 3D morphological processing to remove unwanted regions and produce a Whole Tumor prediction. The supplied ground-truth mask is used for evaluation only and is not passed into the segmentation algorithm.

4.2 Implementation Details

The BraTS2020 training data were used for testing. The evaluation script compares each predicted mask with its corresponding ground-truth mask and computes standard overlap, boundary, and volume metrics. Dice and IoU are calculated as:

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|}, \quad \text{IoU} = \frac{|P \cap G|}{|P \cup G|}, \quad (1)$$

where P denotes the predicted tumor mask and G denotes the ground-truth mask. Precision and recall were also recorded to distinguish false-positive predictions from missed tumor regions. The results were summarized across 20 cases.

5 Results

This section reports the updated evaluation of the current thresholding and morphological-processing pipeline across 20 BraTS2020 Whole Tumor cases. The previous single-slice comparison between the three literature pipelines is not used as the result of this evaluation, because it represents a separate preliminary experiment.

5.1 Quantitative Evaluation

Table 1: Results across 20 BraTS2020 cases for Whole Tumor segmentation.

Metric	Mean $\pm$ SD	Median
Dice	0.665 ± 0.179	0.714
Intersection over Union (IoU)	0.521 ± 0.183	–
Precision	0.787 ± 0.179	–
Recall	0.600 ± 0.211	–
HD95 (mm)	28.12 ± 20.71	25.29
ASSD (mm)	6.96 ± 5.03	5.23
Volume Similarity	0.816 ± 0.127	–

The pooled Dice score was 0.733. A Dice score of at least 0.50 was achieved in 16 of 20 cases (80%), while 11 cases (55%) achieved a Dice score of at least 0.70. The highest Dice scores were 0.888 and 0.850; the lowest were 0.247, 0.342, and 0.405.

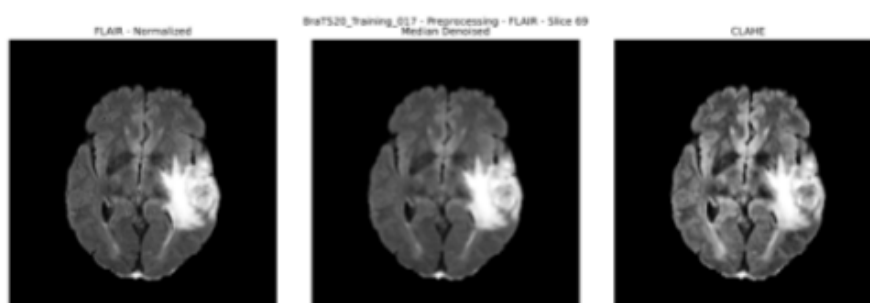


Fig. 1. Example of a preprocessed MRI image.

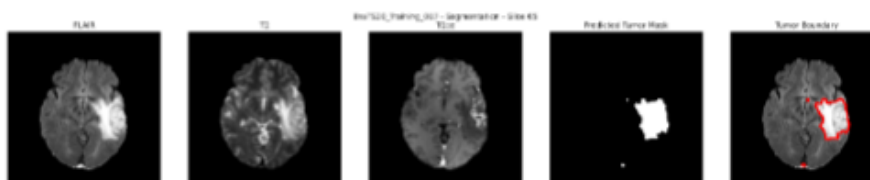


Fig. 2. Example of the predicted tumor segmentation.

Figure 1: Example of MRI preprocessing and predicted tumor segmentation. The upper row shows FLAIR normalization, preprocessing, and CLAHE enhancement; the lower row shows MRI modalities, the predicted tumor mask, and the tumor boundary overlay. This is an illustrative case and not a summary of all 20 evaluated cases.

5.2 Qualitative and Volume Analysis

The predicted masks were generally more precise (mean precision 0.787) than they were complete in terms of recall (mean recall 0.600). This indicates that most of the predicted region coincided with the reference tumor, while some parts of the reference tumor were missing from the predicted region. The case-level review classified 13 cases as under-segmented, 3 as having mixed boundary errors, 2 as over-segmented, and 2 as showing excellent overlap.

The mean signed volume error was approximately 23.3%. Predicted tumor volume was underestimated in 17 cases and overestimated in the remaining 3. Dice overlap also showed a positive association with tumor size (Spearman $\rho = 0.65$), with smaller tumors tending to have lower overlap. These findings are limited to the 20 evaluated cases and do not establish clinical suitability.

5.3 Result Interpretation

The current pipeline can identify tumor regions, but under-segmentation and boundary mismatch remain the main limitations. The difference between precision and recall supports the observation that the pipeline misses parts of some reference tumors. Further work should improve candidate-voxel selection and boundary refinement, followed by evaluation on the same cases and metrics for a fair comparison.

6 Challenges Faced

- Fixed intensity-based rules may not capture tumor intensity variation consistently across patients and MRI modalities.
- Morphological operations can remove unwanted components but may also remove small or thin tumor regions.
- Under-segmentation and boundary errors reduce overlap with the ground-truth masks.
- Smaller tumors tend to have lower Dice overlap, making reliable segmentation more difficult.
- The current evaluation covers 20 cases and the Whole Tumor region only; further validation is required before making broader claims.

7 Plan for Next Week

- Investigate improvements to candidate-voxel selection to reduce under-segmentation.
- Tune thresholding and morphological parameters using validation cases.
- Explore boundary-refinement strategies inspired by the reviewed literature where appropriate.
- Add or improve tumor localization and feature extraction, including area/volume, centroid, and bounding box.
- Re-evaluate changes on the same BraTS2020 cases using Dice, IoU, precision, recall, HD95, ASSD, and volume similarity.
- Maintain a reproducible Python/OpenCV implementation and organize code, outputs, and evaluation results in the project repository.

References

References

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